Day 12 (Oct 31): Revision Sheet

Python dicts and sets | Excel INDEX+MATCH and XLOOKUP missing-ID | loops recall. Fresh recall from memory, no notes.

1. KEY IDEAS

Python: dicts (took longer, more depth)

- **Parallel lists** (names list + marks list) tie data only by position. Sort one list and the pairs break with **no error**. A dict stores key and value together and looks up by key.
- list.index(x) returns the position of the **first** match; a missing value raises **ValueError: 'Zoya' is not in list**. A missing dict key raises **KeyError: 'Zoya'**.
- **get(key, default)** only reads. It returns the value if the key exists, else the default (None if you give none). It never adds the key. The assignment on the left is what creates or updates a key.
- Counting: **counts[w] = counts.get(w, 0) + 1**. get() returns a number (0 for a new word), +1 is added, the assignment stores it. **counts[w] += 1** must read first, so a new word raises KeyError.
- **items()** hands the loop one (key, value) tuple per round, and name, m = unpacks it. A plain *for x in d* gives keys only. Keys must be hashable (a list as key raises TypeError).

Python: sets

- A set is an **unordered collection of unique items**. It is mutable, written with {}, but its items must be hashable. No indexing: s[0] raises TypeError: 'set' object is not subscriptable. {} is an empty dict; use set().
- **&** intersection (in both), **|** union (in either), **-** difference (a - b = in a, not in b; order matters). Converting a list to a set loses counts and order.
- sorted() is a built-in function that takes any iterable and returns a new list. sort() is a list method; a set has no sort() (AttributeError).

Excel

- **MATCH(value, range, 0)** returns the position **inside the range** (from 1), not the sheet row. **INDEX(range, n)** returns the n-th item of that range. MATCH inside INDEX: **=INDEX(B2:B5, MATCH("P103", A2:A5, 0))** = MATCH 3, INDEX 120.
- The 0 means exact match. Leave it out and MATCH defaults to approximate match, which can return a wrong row with no error.
- **XLOOKUP(lookup, lookup_array, return_array, if_not_found)**: exact match by default; the 4th argument shows your text instead of #N/A. A missing ID gives **#N/A** (not #VALUE!).
- IFERROR hides **every** error type; IFNA hides only #N/A. Test a lookup with one ID that exists and one that does not.

2. EXAMPLES TO KNOW

```
counts = {}
for word in ["I", "am", "I"]:
    counts[word] = counts.get(word, 0) + 1     # inside the for loop (4 spaces)
# after each round: {'I': 1}  ->  {'I': 1, 'am': 1}  ->  {'I': 2, 'am': 1}
```

```
marks = {"Asha": 80, "Ravi": 65, "Meena": 92}
marks["Zoya"]        # KeyError: 'Zoya'
marks.get("Zoya", 0)  # 0   (nothing is stored)
a = {1, 2, 3}; b = {2, 3, 4}
a & b -> {2, 3}    a | b -> {1, 2, 3, 4}    a - b -> {1}    sorted(a & b) -> [2, 3]
```

3. MY MISTAKES AND THE FIX

What I said or did	The fix
MATCH("P103") = 4; INDEX(B2:B5, 3) = 75	Position inside the range: MATCH = 3, INDEX = 120. Not the sheet row.
XLOOKUP missing gives #VALUE!	#N/A means not found. #VALUE! is a wrong argument type.
Dict counting works 'because dict remembers order'	It works by key lookup: get() reads the current count, the assignment stores count + 1.
get() returns the key; 'a has no value'	get() returns the value or the default and stores nothing. A missing key does not exist at all.
{'a': 1} + 1 gives {'a': 2}	get() returns the number 1; 1 + 1 = 2 is stored. dict + int is a TypeError.
Wrote only the changed pair after a round	Write the whole dict after every round. Keys are case-sensitive: 'I' is not 'i'.
Set is immutable, shown with [ ]	A set is mutable, shown with { }. Only its items must be hashable.
a - b = {4}	a - b keeps items of a not in b: {1}. {4} is b - a.
sorted is a list method	sorted() is a built-in function. sort() is the list method; a set has no sort().
Merge commit when 'different files change'	Merge commit comes from diverged branches (both have new commits), not from which files change.

4. HOW I IMPROVED

- Counting loop: stuck at round 1, then traced all three rounds correctly once I split it into steps.
- INDEX+MATCH: wrong numbers at first, then built the correct formula and tested both a found and a missing ID in Excel.
- Problems 1 to 3 (word count, topper, common elements) passed; Problem 1 needed a hint after 20 minutes.

5. PRACTICE TO RETRY (no solutions)

- Trace counts.get on "to be or not to be": write the whole dict after every round.
- Topper with a tie on 92: print **all** toppers. Then say what changes if you use >= instead of >.
- l1 = [4, 7, 9, 7, 2], l2 = [7, 2, 8, 2, 1]: print the values only in l1, sorted, and how many.
- Price table (IDs A2:A5): XLOOKUP with the message "Not listed", and INDEX+MATCH with IFNA. Test P104 and P999.
- Predict the exact error name and message: d["x"] on a missing key, s[0] on a set, [1,2] & [2,3] on two lists.

6. RECALL QUESTIONS (answer from memory, no notes)

- What does get(key, 0) return for a missing key, and does it add the key?
- Why does counts[w] += 1 fail on a new word, and with which error?
- What does items() give the loop each round? What does a plain loop over a dict give?
- What is a set? What do &, | and - do? What do you lose converting a list to a set?
- MATCH returns what, and why is it not the sheet row? Write INDEX+MATCH for P103 and give both values.
- XLOOKUP missing ID: which error, which argument gives a message, and what does IFERROR risk?

7. OPEN ITEMS

- Hardest-problem redo from a blank file: lists/tuples (Oct 30) and dicts/sets (Oct 31) not done.
- Word counter handles only commas: "!" and "." stay glued; strip each word after split().
- git push from Colab with a token not retested. Fast-forward vs merge commit: retest again.